Asymmetric Bidirectional Fusion Network for Remote Sensing Pansharpening

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***Abstract*— Pansharpening aims to generate a high-resolution multispectral (HR-MS) image given a paired panchromatic (PAN) image and low-resolution multispectral (LR-MS) image. Though the existing pansharpening methods have made remarkable progress, the fusion pipeline does not fully adapt to the distinct characteristics of the PAN and LR-MS images. In this article, to fully exploit the complementary modality of the two images, we propose a novel and efficient asymmetric bidirectional fusion network (ABFNet). The ABFNet consists of the two customized fusion modules with asymmetric architectures, which aim to reinforce the PAN and LR-MS images, respectively. Specifically, the spectral colorization (SC) module recalibrates the scale and bias of the PAN features using weights generated by the LR-MS features, which aims to inject spectral information into the PAN features without breaking their spatial continuity. To transfer spatial details from the PAN features into the LR-MS features, the spatial restoration codebook (SRC) module refines the LR-MS features with point-to-point restoration codebooks learned from the PAN features. By incorporating the two modules in multiple stages, ABFNet enjoys a high capability for capturing both spectral and spatial dependencies. Extensive experiments over multiple satellite datasets demonstrate the effectiveness of the proposed methods.**

***Index Terms*— Asymmetric modules, bidirectional learning, image fusion, pansharpening.**

I. Introduction

NOWADAYS, high-resolution multispectral (HR-MS) images are essential for many remote sensing tasks (i.e., object recognition [1], [2], semantic segmentation [3], [4], and mapping services [5], [6]), since HR-MS images can provide more accurate information of land covers and detailed structures of object targets. However, due to technological and physical constraints, it is hard for remote sensing satellites to capture HR-MS images directly. As a compromise, these satellites are equipped with both multispectral and panchromatic (PAN) sensors that capture low-resolution multispectral (LR-MS) and high-resolution PAN images, respectively. Under this condition, to generate HR-MS images for downstream tasks, an effective solution is fusing the PAN and LR-MS images. The fusion process, called pansharpening, aims to integrate the spectral property of the LR-MS image and spatial details of the PAN image.

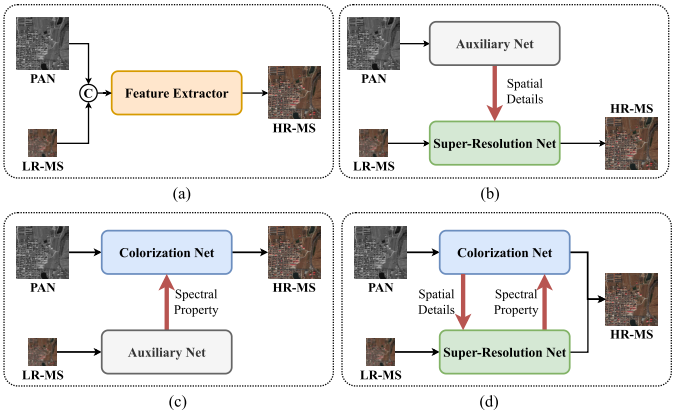


Fig. 1. Fusion pipelines of different pansharpening methods. (a) Concatenation-based methods. (b) Super-resolution-based methods. (c) Colorization-based methods. (d) Bidirectional learning-based methods. The bidirectional learning-based methods unify (b) and (c) by enabling bidirectional information interaction.

Over the past few decades, many research efforts have been devoted to improve the visual quality of the pansharpened HR-MS image. According to different fusion pipelines, they can be categorized into concatenation-based methods, super-resolution-based methods, and colorization-based methods. Concatenation-based methods [7], [8], [9], [10], [11], as presented in Fig. 1(a), fuse the two images by directly concatenating the heterogeneous images or features; then, they rely on powerful feature extractors, such as neural networks, to learn the mixed features automatically. Super-resolution-based methods treat the pansharpening task as a super-resolution problem. They adopt a super-resolution net to generate the HR-MS images by super-resolving LR-MS images, and PAN images are used to provide spatial details for LR-MS features by an auxiliary net, as shown in Fig. 1(b). Traditional methods use different analysis tools (e.g., intensity–hue–saturation (IHS) [12], [13], principal component analysis (PCA) [14], [15], and wavelet transform [16], [17]) to compute injection gains that control the amount of spatial information injected from PAN images to LR-MS images. Recently, some works use deep-learning networks and leverage attention module [18], [19] or adaptive convolution kernels [20], [21] to transfer spatial details from PAN images to LR-MS images in the feature domain. Colorization-based

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methods presented in Fig. 1(c) view the pansharpening task as a colorization problem. They propose a colorization net to colorize the PAN image under the guidance of spectral properties from the LR-MS image. Specifically, they conducted spectral colorization (SC) at frequency domain [22], [23] or adjusted the distribution of the PAN features by normalization modules [24].

Overall, concatenation-based methods implicitly construct the relationship in the neural networks, which neglect the distinct property of the PAN and LR-MS images. Super-resolution and colorization-based methods design explicit fusion pipelines to incorporate information from PAN to LR-MS image and LR-MS to PAN image, respectively. Due to the effective feature fusion pipelines, they demonstrate better performance than the concatenation-based method. However, they focus on enhancing only one of the two images (LR-MS image in the super-resolution-based methods and PAN image in the colorization-based methods), so the quality of the pansharpened HR-MS image is highly dependent on this image. When the image suffers from severe degradation, they will fail to generate plausible HR-MS images. Meanwhile, another image is only used to provide auxiliary information, which neglects its potential of serving as the basis to reconstruct the HR-MS image. To relieve the limitations of these methods and fully leverage the PAN and LR-MS images, we resort to the bidirectional learning paradigm, as presented in Fig. 1(d), which combines the super-resolution and colorization nets in a unified framework. Bidirectional learning-based methods [25], [26], [27], [28], [29] reinforce the PAN and LR-MS features by constructing bidirectional interaction connections. For example, gradient projection based pansharpening neural network (GPPNN) [25] sequentially optimizes the PAN and LR-MS features in two optimization modules. Bidirectional pyramid network (BPN) [26] and triple-double convolutional neural network (TDNet) [27] extend bidirectional fusion into multiscale features. Zhu and Zhang [28] use saliency maps to guide the bidirectional fusion process, and Zhou et al. [29] maximize the mutual information between the PAN and LR-MS features in a bidirectional way. Though they construct bidirectional learning pipeline to fuse the two images, they focus on adopting it with the existing architectures, and the fusion modules for PAN and LR-MS features often share symmetric architectures, which is not optimal for the two images with different modalities. To further adapt to the distinct characteristics of the PAN and LR-MS images, we propose an asymmetric bidirectional fusion network (ABFNet) for remote sensing pansharpening. The ABFNet consists of two fusion modules with asymmetric architectures, which aim to reduce redundant features and improve fusion efficiency.

To expand the PAN image into multiple spectral bands, spectral information needs to be injected from the LR-MS image. At the same time, the spatial structure of the PAN image should be kept to maintain its high spatial resolution. The two requirements are concurrent to the idea of style transfer methods [30], [31], which aims to change the style of the target image meanwhile keeping its content unchanged. Inspired by these methods, we propose the SC module to perform the LR-MS to PAN fusion. Specifically, the SC module only resets the scale and bias for the normalized PAN features, where the scale and bias are conditioned on the corresponding LR-MS features. In this way, the SC module only changes the statistical information of the PAN features, so it can maintain the semantic information of the PAN features. On the other hand, to super-resolve the LR-MS image into the corresponding HR-MS image, we propose the spatial restoration codebook (SRC) module that transfers spatial details from the PAN image to the LR-MS image and also preserves the original spectral property of the LR-MS image. The SRC module refines the LR-MS features in a point-to-point manner under the guidance of the PAN features. In the SRC modules, the LR-MS feature points are viewed as code words that embed compact spectral information and deteriorate spatial details. To transfer spatial details, we convert the PAN feature points into restoration codebooks for the LR-MS code words. By customizing the restoration codebook for different LR-MS features, the SRC module can adaptively strengthen them concerning their contents and semantic information. Compared with methods [10], [32], [33] that use static convolution kernels, which are agnostic to feature content, the SRC module can effectively aggregate the spatial details for the LR-MS features. The SC and SRC modules comprise our proposed ABFNet, which intends to divide the pansharpening into spectral and spatial fusion. By incorporating the two modules in multiple stages, ABFNet enjoys a high capability for capturing both spectral and spatial dependencies. Overall, our main contributions are summarized as follows.

1) We leverage the bidirectional learning pipeline and propose the ABFNet for remote sensing pansharpening. The ABFNet enables interaction between the PAN and LR-MS image pair, which can fully exploit the representation potential of complementary information embodied in these two images.
2) We propose the SC and SRC modules that conduct LR-MS to PAN fusion and PAN to LR-MS fusion, respectively. The two fusion modules consider the distinct properties of the two inputs and further facilitate the bidirectional learning pipeline.
3) Extensive experiments on three datasets demonstrate the superior performance of the proposed ABFNet.

The rest of this article is organized as follows. Section II reviews related works. Section III introduces our proposed method in detail. Section IV presents experimental results and analysis. Section VI concludes this article.

II. Related Work

A. Traditional Pansharpening Methods

Traditional pansharpening methods can be categorized into three ways: component substitution-based (CS) methods, multiresolution analysis-based (MRA) methods, and variational optimization-based (VO) methods. The CS methods first transform the LR-MS into a new space and substitute some components with the PAN slice. The inverse transformation is performed to reconstruct the predicted pansharpened image. Some methods [12], [13] converted the PAN and LR-MS images into IHS and matched the histogram of the PAN image with the histogram of the LR-MS image. Some other methods [14], [15] substituted the principal vectors of the PAN image with the LR-MS image. Other methods adopted

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Gram–Schmidt (GS) [34], [35] to decompose the two images and substitute the coefficients. These methods conduct substitution in the image domain or shallow features, leading to severe artifacts and content discontinuity. The MRA methods inject high spatial details from multiresolution PAN images into LR-MS images. Some methods [16], [17] resorted to MRA tools, such as wavelet transform and Gaussian pyramid, to generate multiscale PAN images. Some methods adopted modulation transfer function (MTF [36], [37]) to calculate the injection gains. Other methods used variation factors [38] and scale regression [39] to improve the performance. The quality of the generated image benefits from multiscale processing, but the fusion in the image domain is still not ideal. The VO methods [40], [40], [41], [42], [43] aim to find an optimized solution using human-designed assumptions and regularization. Such assumptions and regularization work well in simple conditions, but when faced with complex scenes, they may not generalize well. In general, due to limited model capacity, the performance of traditional pansharpening methods is limited and often suffers from spatial or spectral distortions.

B. Deep-Learning-Based Pansharpening Methods

With the development of deep-learning techniques, various deep-learning-based methods have been proposed to tackle the multispectral pansharpening problem. CNN-based pansharpening (PNN) [7] is the first work to apply CNN into remote sensing pansharpening, which used three convolutional layers to fuse the PAN and LR-MS images. The global residual connection was used in [9], [44], and [45] to ease the HR-MS regression by only learning the residual part. Multiscale and multidepth convolutional neural network (MSDCNN) [10] tried aggregating multiscale features by adopting multiple convolution layers with different kernel settings. Bilateral activation mechanism based network (BAM) [11] proposed a bilateral activation mechanism to offer an efficient feature extraction model, which exploited the negative part of the features. Saliency cascade convolutional neural network for pansharpening (SC-PNN) [46] adopted saliency cascade convolution to progressively refine the HR-MS image. Vision Transformer is explored in [32], [47], and [48] to capture global information of the mixed input. These works concatenate the PAN image and LR-MS image as network input and focus on exploring different network structures. Recently, some works resort to explicit fusion pipelines to generate plausible HR-MS images efficiently. NormNet [24] progressively transferred the spectral information into the PAN feature using normalization modules. Source-adaptive discriminative kernels based network (ADKNet) [20] generates source-adaptive convolutional kernel using the PAN and MS features. PanColorGAN [49] adopted an adversarial generative network (GAN) to control the spectral injection from LR-MS to PAN images. Cross attention is used in [19] to construct the interactions between the PAN features and the LR-MS features. Chen et al. [50] proposed an iterative super-resolution network for learning cross-scale interactions. However, these works take one of the PAN and LR-MS images as the basis for the HR-MS restoration, making another image underexploited. The results produced are often biased toward the basic image. In this work, different from the above networks, we propose a model to generate the HR-MS image based on the two images.

C. Bidirectional Learning Methods

Compared with unidirectional methods, bidirectional learning can capture more contextual information and enhance the performance of various tasks. Bidirectional long short-term memory (BiLSTM) network [51] and bidirectional recurrent neural networks (BiRNNs) [52] can offer better predictions than LSTM and RNN in tasks, such as speech recognition [53], time-series forecasting [54], and sentimental analysis [55], since it can aggregate information from both directions: from left to right and from right to left. In [56], bidirectional learning was used to construct the forward and backward feature connections for scene flow estimation, which enhanced the estimation performance. Li et al. [57] adopted the bidirectional learning for domain adaption of semantic segmentation to improve the segmentation model and auxiliary image translation model simultaneously. Luo et al. [58] proposed a bidirectional vision language to fully explore the relationship between the vision and language modality, which shows superior performance on visually rich document understanding. In remote sensing tasks, bidirectional learning has also been explored to better process the remote sensing images. Sun et al. [59] proposed a gated bidirectional network for scene classification. Bidirectional word vectors were proposed in [60] to match the contextual information and image pixel. Abdullah et al. [61] explored a bidirectional triplet network to match text with remote-sensing images. Inspired by the above methods, which adopt bidirectional learning to construct the relationship between two modalities, we use bidirectional learning to transfer the complementary information of the PAN and LR-MS image. Several methods have also adopted bidirectional learning to promote the performance of remote sensing pansharpening. A bidirectional recurrent pansharpening network (ArbRPN) [62] adopted the bidirectional mechanism to generate the HR-MS image with arbitrary numbers of bands, and they focus on exploring the bidirectional relationship between different bands. Zhang et al. [26] proposed a BPN to pansharpen the PAN and LR-MS images from coarse to fine scale, but they simply added the features to transfer information. Joint visual saliency analysis and parallel bidirectional network (JSPBN) [28] extended the BPN by using the saliency map as a fusion guidance, but it took too many resources to calculate the saliency maps. Different from theirs, we aim to build a light and efficient network, and we designed asymmetric fusion modules to explicitly integrate the complementary information, rather than solely adding them.

III. Proposed Method

In this section, we will elaborate on the proposed ABFNet.

A. Problem Definition

Pansharpening aims to reconstruct HR-MS images by fusing the corresponding PAN and LR-MS images. The degradation processes [63] that relate the three images can be expressed by the following equations:

$$\begin{aligned} P &= Hw + n_p \\ L &= (H \otimes k) \downarrow_4 + n_L. \end{aligned} \tag{1}$$

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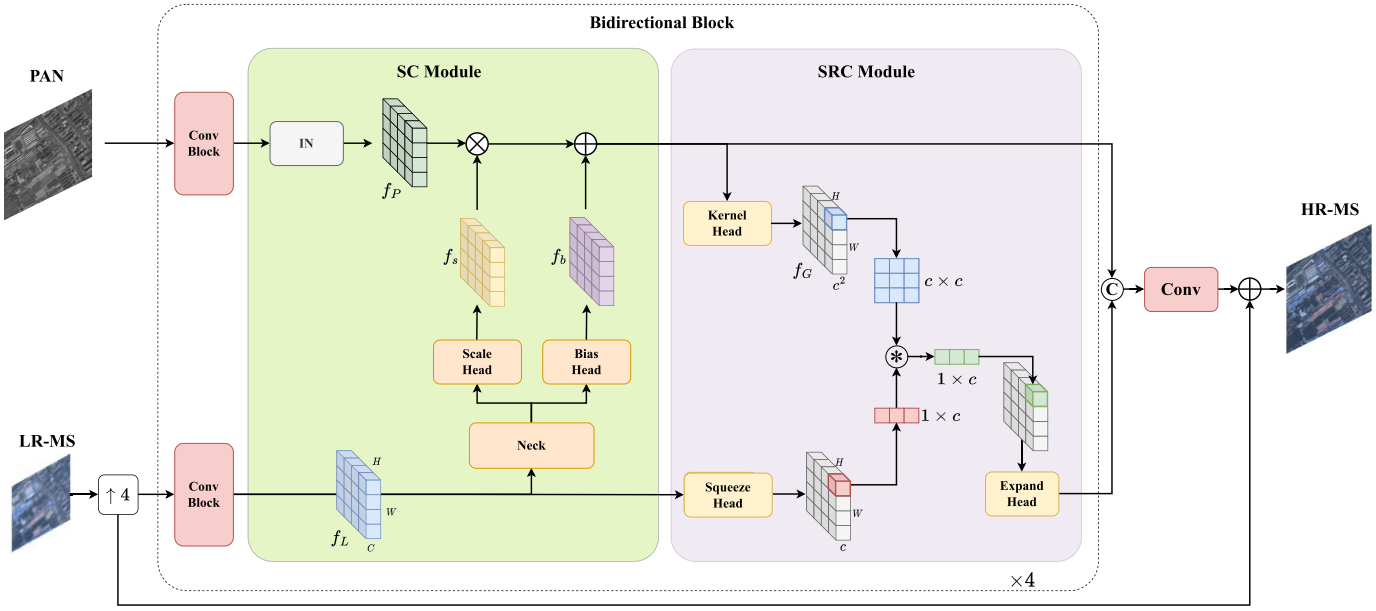


Fig. 2. Overall structure of the proposed method. The BBs sequentially extract and fuse the PAN and LR-MS features. The SC module aims to fuse spectral responses into the PAN features and colorize them into multiple spectral bands. The SRC module aims to improve the spatial resolution of the LR-MS features by using PAN features as a guidance.

P, L, and H are the PAN, LR-MS, and HR-MS images, respectively. w is the spectral band reduction kernel of shape $\mathbb{R}^{C\times 1}$, which produces the PAN image by reducing the spectral bands of HR-MS images from C into 1. k is the spatial blurring kernel, $\otimes$ represents the convolution operation, and $\downarrow_4$ denotes downsampling operation by a factor of 4. By blurring and downsampling the HR-MS image, we will get the LR-MS image. n_P and n_L are the additive noise for PAN and LR-MS images, respectively. In this way, the image shape of LR-MS, PAN, and HR-MS is $\mathbb{R}^{H/4\times W/4\times C}$, $\mathbb{R}^{H\times W\times 1}$, and $\mathbb{R}^{H\times W\times C}$, where H and W are the height and width of the HR-MS image, respectively, and C is the number of spectral bands.

Pansharpening can be viewed as the inverse problem of the degradation processes, which fuses the PAN and LR-MS images to restore the HR-MS image. The process of pansharpening can be defined as follows:

$$H = \mathbb{N}_\theta(P, L) \tag{2}$$

where $\mathbb{N}$ denotes pansharpening network and θ denotes its parameters. By integrating the PAN and LR-MS images, the pansharpening network can produce realistic HR-MS images that preserve both the spectral property of the LR-MS image and the spatial information of the PAN images.

B. Model Architecture

The overall architecture of the proposed ABFNet is shown in Fig. 2. The ABFNet first upsamples the LR-MS by a factor of 4 as follows:

$$\hat{L} = \text{bicubic}(L, 4) \tag{3}$$

where L is the LR-MS image, $\hat{L}$ is the upsampled LR-MS image, bicubic denotes the bicubic interpolation methods, and 4 is the upsampling factor.

The upsampling operation enables the upsampled LR-MS image $\hat{L}$ to share the same spatial dimension with the PAN image, which makes it possible to conduct feature fusion in a point-to-point manner and helps align the receptive fields of the PAN and LR-MS features in the subsequent stages.

The ABFNet then uses four bidirectional blocks (BBs) to extract and fuse spatial details and spectral properties of the PAN and LR-MS features. The four BBs are stacked sequentially, allowing them to gradually incorporate textual details and semantic information from shallow to deep levels, which can be represented as follows:

$$f_P^i, f_L^i = \text{BB}_i\left(f_P^{i-1}, f_L^{i-1}\right) \tag{4}$$

where BB_i denotes the ith BB, and f_P^i and f_L^i denote the output PAN and LR-MS features of the ith BB. Notably, the input features of the first BB are the PAN image P and upsampled LR-MS image $\hat{L}$, which means $f_P^0 = P$ and $f_L^0 = \hat{L}$.

In each BB, as shown in the dashed box, bidirectional fusion modules are constructed to fuse the PAN and LR-MS features. Specifically, the SC module aims to strengthen the PAN features with spectral properties extracted from the LR-MS features. The SRC module aims to provide spatial details from the PAN features to super-resolve the LR-MS features. The SC and SRC modules use different fusion pipelines to aggregate spectral and spatial information for the PAN and LR-MS features, which can better adapt to their distinct characteristics. The optimization process of the SC and SRC modules can be summarized as follows:

$$\begin{aligned} f_P &= \text{SC}(f_P, f_L) \\ f_L &= \text{SRC}(f_P, f_L). \end{aligned} \tag{5}$$

Besides the feature fusion modules (SC and SRC), each BB also consists of two convolution blocks (ConvBlocks), which

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aim to extract deeper PAN and LR-MS features as follows:

$$f_P = \text{ConvBlock}(f_P)$$
$$f_L = \text{ConvBlock}(f_L). \tag{6}$$

To keep the model complexity in a reasonable level, the channel dimension of all features is set to 32, unless otherwise stated. After feature extraction and fusion of the BBs, the reinforced PAN and LR-MS features can be viewed as two candidates for the HR-MS features, since they possess both multispectral properties and high spatial resolution. To robustly and faithfully reconstruct the HR-MS image, the PAN and LR-MS features are combined and fed into the decoder. The decoder aims to exploit the mutual information of the two features and filter out noisy interference brought by a single modality. Specifically, we concatenate the PAN and LR-MS features together and pass them through a convolution layer to predict the residual image, which measures the difference between the ground-truth HR-MS image and the upsampled LR-MS image. By predicting the residual image rather than the full HR-MS image, the network can converge faster and achieve better performance, as evidenced by previous methods in remote sensing pansharpening [9], [44], [45]. The residual image is added to the upsampled LR-MS image $\hat{L}$ to produce the HR-MS image. The reconstruction process can be represented as follows:

$$H = \text{Conv}\left(\left[f_P^4, f_L^4\right], \text{kernel} = 1\right) + \hat{L} \tag{7}$$

where Conv is the convolution layer and kernel = 1 means the kernel size of the convolution layer is 1×1. f_P^4 and f_L^4 are the output PAN and LR-MS features of the fourth (final) BB, respectively. $[f_P^4, f_L^4]$ denotes the concatenation of the PAN and LR-MS features along the feature dimension.

C. SC Module

Concurrent to style transfer works [30], [31], which aims to inject style attributes into the target image and keep the contents of the target image unchanged, the SC module of the ABFNet aims to inject spectral property into the PAN image and keep the spatial details of the PAN image. Based on the observation, the SC module adopts the same adaptive normalization mechanism, which resets the scale and bias of the PAN features based on the feature response learned from the LR-MS features. The adaptive normalization mechanism can effectively transfer the spectral property from the LR-MS features to the PAN features without destroying their spatial continuity.

Specifically, we first use the instance normalization (IN) layer [64] to remove noise variations in the PAN features and standardize their intensity distribution. The IN layer normalizes PAN features $f_P \in \mathbb{R}^{C\times H\times W}$ with their per-dim mean μ and standard deviation σ as follows:

$$f_P = \frac{f_P - \mu}{\sqrt{\sigma^2 + \epsilon}} \tag{8}$$

where ϵ is a small constant that avoids the denominator equals zero.

Given a spatial location i and j, the per-dim mean μ_{ij} measures the average intensity along the channel dimension of the feature point f_{ij} as follows:

$$\mu_{ij} = \frac{1}{C}\sum_{k=0}^{C} f_{ij}^k \tag{9}$$

where C is the number of channels.

Meanwhile, the per-dim standard deviation σ_{ij} measures the intensity variations along the channel dimension of the feature point f_{ij} as follows:

$$\sigma_{ij} = \sqrt{\frac{\sum_{k=0}^{C}\left(f_{ij}^k - \mu_{ij}\right)^2}{C}} \tag{10}$$

where μ_{ij} is the per-dim mean calculated above and C is the number of channels.

Then, the normalized PAN features f_P are recalibrated by scale features f_s and bias features f_b derived from the LR-MS features f_L as follows:

$$f_P = f_s \times f_P + f_b. \tag{11}$$

In this way, the scale and bias features stretch and offset the distribution of the PAN features toward the distribution of the LR-MS features. Moreover, the SC module only adjusts the statistical information of the PAN features, so the spatial contents of the PAN features can be preserved contact.

To generate the scale and bias features (f_s and f_b), a convolution neck is adopted to extract preliminary features from the LR-MS features f_L as follows:

$$f = \text{ReLU}(\text{Conv}(f_L, \text{kernel} = 1) \tag{12}$$

where Conv is the convolution layer with a kernel size of 1×1 and ReLU is the activation function to introduce nonlinear characteristics.

Then, the scale head adopts a convolution layer with a kernel size of 1×1 to generate the scale features $f_s \in \mathbb{R}^{C\times H\times W}$, while the bias head also adopts a convolution layer with a kernel size of 1×1 to produce the bias features $f_b \in \mathbb{R}^{C\times H\times W}$. The generation process of the scale and bias features is as follows:

$$f_s = \text{Conv}(f, \text{kernel} = 1) \tag{13}$$
$$f_b = \text{Conv}(f, \text{kernel} = 1). \tag{14}$$

The scale and bias feature generator only consists of three convolution layers, which makes the generation process light and efficient.

D. SRC Module

To transfer spatial details from the PAN branch to the LR-MS branch, we will first delve into the degradation process from the HR-MS image to the LR-MS image in (1). The LR-MS image is generated by blurring and downsampling the HR-MS image with the spatial blurring kernel k and downsampling operation $\downarrow$, so to reconstruct the HR-MS image from the LR-MS image, we need to perform the inverse operation of downsampling and blurring, namely, upsampling

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and restoration. For the ABFNet, we use the bicubic interpolation in (3) to represent the upsampling operation, and the SRC module is used to perform restoration for the LR-MS features.

Specifically, the LR-MS features f_L are compressed into a lower dimensional subspace to reduce computation and memory costs. The squeeze head uses a 1×1 convolution layer to reduce the number of channels from C to c. In the experiment, c is a quarter of C, which is set empirically. The squeeze head is set as follows:

$$f_L = \text{Conv}(f_L, \text{kernel} = 1) \tag{15}$$

where Conv denotes the convolution layer and kernel $=$ 1 means the kernel size is 1×1.

With LR-MS feature dim set to c, we generate the corresponding SRCs f_G from the PAN features f_P with the kernel head as follows:

$$f_G = \text{Conv}(f_P, \text{kernel} = 1) \tag{16}$$

where the kernel head consists of a convolution layer with a kernel size of 1×1. It produces the SRC features $f_G \in \mathbb{R}^{c^2 \times H \times W}$ with c^2 channels and the same spatial size as the LR-MS features. By transforming the PAN features into the restoration codebooks, we aim to embed representative spatial details of the corresponding areas and targets.

After the feature reduction and codebook generation, the LR-MS features are refined using the restoration codebook in a point-to-point manner. As presented in Fig. 2, given the red feature point $f_L^{ij} \in \mathbb{R}^{1 \times c}$ of squeezed LR-MS features at spatial location i and j, the corresponding blue codebook $f_G^{ij} \in \mathbb{R}^{1 \times c^2}$ is first reshaped into a $c \times c$ matrix, and the LR-MS feature point f_L^{ij} is reinforced by multiplying the restoration codebook f_G^{ij} as follows:

$$f_L^{ij} = f_L^{ij} * F_G^{ij} \tag{17}$$

where $*$ denotes the matrix multiplication. This operation produces a new feature point $f_L^{ij} \in \mathbb{R}^{1 \times c}$ highlighted in green, and by performing the restoration process across different spatial feature points, we will get the refined LR-MS features f_L.

After the restoration operation, the expand head in the bottom projects LR-MS features back into its original space with C channels. It uses another 1×1 convolution layer to restore the dimensionality of the LR-MS features as follows:

$$f_L = \text{Conv}(f_L, \text{kernel} = 1). \tag{18}$$

Rather than directly learning the spatial restoration matrix conditioned on neighborhood pixels for the LR-MS image, the SRC module performs the restoration in the feature domain with gradually increasing receptive fields and learns restoration codebooks conditioned on the PAN features to refine the LR-MS features. In this way, the SRC module can effectively improve the spatial resolution of LR-MS images by leveraging the PAN features as a source of guidance. It can capture local spatial patterns, which adapt to different feature points, and preserve the spectral information of the LR-MS images without introducing spectral distortion or artifacts.

E. ConvBlock

Feature fusion at SC and SRC modules in a point-to-point manner only matches the features at the same spatial location, which may ignore some useful information from the surrounding features, so the ConvBlock is a component that enhances the feature fusion process by considering the neighborhood features of the PAN and LR-MS images. By aggregating the local features of the PAN and LR-MS features separately, the ConvBlock can capture more spatial details and context information, which improves the fusion quality and accuracy.

Specifically, the ConvBlock consists of a 3×3 convolution layer, a batch-norm (BN) layer, and a ReLU layer. The feature extraction pipeline of the ConvBlock is as follows:

$$f = \text{ReLU}(\text{BN}(\text{Conv}(f, \text{kernel} = 3)) \tag{19}$$

where f denotes PAN or LR-MS features. In the first BB, the two ConvBlocks are used to extract preliminary features from the PAN and upsampled LR-MS images, respectively. In the remaining BBs, the ConvBlocks aims to refine the fusion results of the SC and SRC modules and expand the receptive fields for the subsequent SC and SRC modules.

F. Loss Setting

Following previous works [9], [65] in remote sensing pansharpening, the training loss function is set as the $L1$ loss due to its simplicity and effectiveness. The $L1$ loss calculates the absolute difference between the reconstruction HR-MS image and the ground-truth HR-MS image as follows:

$$\text{Loss} = \|\mathbb{N}_\theta(P, L) - H\|_1 \tag{20}$$

where P and L are the paired PAN and LR-MS images, respectively. H is the corresponding ground-truth HR-MS image, $\mathbb{N}_\theta$ represents the network and its parameters, and $\mathbb{N}_\theta(P, L)$ denotes the predicted HR-MS of the network. During training, the network optimizes its parameters θ by minimizing the $L1$ loss, and during testing, it tries to produce a reconstructed HR-MS image that is close to H in terms of the $L1$ loss.

IV. Experiments

A. Datasets

To evaluate the performance of our method and compare it with other pansharpening methods, we use image pairs collected from three different satellites: WorldView-3 (WV-3), GaoFen-2 (GF-2), and QuickBird (QB). The three satellites have different spectral bands, resolutions, and pixel value ranges. Details of the three satellite datasets are presented in Table I. Experiments are conducted on two types of resolution: reduced resolution (RR) and full resolution (FR). RR is generated under the Wald protocol [69], which downsamples the original PAN and MS images by a factor of 4 to simulate the PAN and LR-MS image pairs, and the original MS images are used as the ground-truth targets to supervise the network training and evaluation. The image pairs are split into training and testing sets. FR uses original MS and PAN images directly and aims to evaluate the generalization ability of different methods, which are trained on the RR training sets. Details of the three satellites are given below.

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TABLE I
DETAILS OF THE DATASETS USED IN THE EXPERIMENT

Sensor	MS bands	Bit Depth	Reduced Resolution			Full Resolution	
			Resolution	Training	Testing	Resolution	Testing
GaoFen-2	4	10	PAN:3.2m MS:12.8m	Patches:5827 PAN:256 × 256 × 1 MS:64 × 64 × 4	Patches:680 PAN:256 × 256 × 1 MS:64 × 64 × 4	PAN:0.8m MS:3.2m	Patches:170 PAN:1024 × 1024 × 1 MS:256 × 256 × 4
QuickBird	4	11	PAN:2.4m MS:9.6m	Patches:196 PAN:256 × 256 × 1 MS:64 × 64 × 4	Patches:60 PAN:256 × 256 × 1 MS:64 × 64 × 4	PAN:0.6m MS:2.5m	Patches:21 PAN:1024 × 1024 × 1 MS:256 × 256 × 4
WorldView-3	8	11	PAN:1.6m MS:6.4m	Patches:311 PAN:256 × 256 × 1 MS:64 × 64 × 8	Patches:238 PAN:256 × 256 × 1 MS:64 × 64 × 8	PAN:0.4m MS:1.6m	Patches:63 PAN:1024 × 1024 × 1 MS:256 × 256 × 8

1) GaoFen-2: It is a high-resolution optical Earth observation satellite. The resolution of its PAN and multispectral imagery is 0.8 and 3.2 m. The MS imagery has four spectral bands, and the bit depth of pixels in the two imagery is 10, which means the maximum pixel value is 1024. The dataset contains six PAN and MS imagery pairs. We randomly select five imagery pairs as the training set and select the remaining imagery pair as the testing set. For RR, we crop the PAN imagery into 256 × 256 × 1 patches and crop the MS imagery into 64 × 64 × 4 patches. The total patch number is 5872 and 680 for training and testing, respectively. For FR, we crop the PAN imagery into 1024 × 1024 × 1 patches and crop the MS imagery into 256 × 256 × 4 patches, resulting in 170 patches.

2) WorldView-3: It can capture PAN imagery of 0.4-m resolution and a multispectral imagery of 1.6-m resolution with eight spectral bands. The bit depth of pixels in the two images is 11, which means the maximum pixel value is 2048. The dataset contains three pairs of PAN and MS imagery. We randomly select two imagery pairs as the training set and select the remaining pair as the testing set. For RR, we crop the PAN imagery into 256 × 256 × 1 patches and crop the MS imagery into 64 × 64 × 8 patches. The total patch number is 311 and 238 for training and testing, respectively. For FR, we crop the PAN imagery into 1024 × 1024 × 1 patches and crop the MS imagery into 256 × 256 × 8 patches, resulting in 63 patches.

3) QuickBird: It can capture PAN imagery of 0.6-m resolution and a multispectral imagery of 2.4-m resolution with eight spectral bands. The bit depth of pixels in the two images is 11, which means the maximum pixel value is 2048. The dataset contains four pairs of PAN and MS imagery. We randomly select three imagery pairs as the training set and select the remaining pair as the testing set. For RR, we crop the PAN imagery into 256 × 256 × 1 patches and crop the MS imagery into 64 × 64 × 4 patches. The total patch number is 196 and 60 for training and testing, respectively. For FR, we crop the PAN imagery into 1024 × 1024 × 1 patches and crop the MS imagery into 256 × 256 × 4 patches, resulting in 21 patches.

B. Implementation Details

We compare our ABFNet with both traditional pansharpening methods and deep-learning-based methods. Traditional methods include CS methods: IHS [12], Brovey [66], GS [34], and PCA [14]; and MRA methods: smoothing filter-based intensity modulation (SFIM) [67] and Wavelet [17]. Deep-learning-based methods includes concatenation-based methods: PNN [7], PanNet [9], MSDCNN [10], and BAM [11]; super-resolution-based methods: super-resolution-guided progressive pansharpening neural network (SRPPNN) [68]; colorlization-based methods: NormNet [24]; and bidirectional learning-based methods: GPPNN [25].

For traditional methods, we directly evaluate them on the RR and FR testing sets. For deep-learning methods, we retrain them using RR training sets of the three satellite datasets, and then, these networks are evaluated on the RR and FR testing sets directly without fine-tuning or adaption. For fair comparisons, all networks are trained under the same hyper-parameters. The initial learning rate is 0.0002, and the batch size is set to 8. The number of training epochs for each satellite dataset is set according to their size and complexity. Specifically, for the WV-3 dataset, networks are trained for 500 epochs, and the learning rate is reduced by half every 100 epochs. For the GF-2 dataset, networks are trained for 100 epochs, and the learning rate is reduced by half every 20 epochs. For the QB dataset, networks are trained for 1000 epochs, and the learning rate is reduced by half every 200 epochs. Experiments are conducted on a single NVIDIA 3090Ti GPU under PyTorch 1.12.

C. Evaluation Metrics

To measure the quality of the pansharpened images, we use two sets of metrics: RR metrics and FR metrics for corresponding datasets. The RR metrics are used to compare the network output with the ground-truth MS images in a pixelwise manner. The FR metrics are used to assess the network performance on the original PAN and MS images without ground truth.

1) RR Metrics: They include four commonly used metrics: cross correlation (CC), spectral angle mapping (SAM) [70], errur relative globale adimensionnelle desynthese (ERGAS) [71], and peak signal-to-noise ratio (PSNR).

1) **CC** measures the correlation between the spectral bands of the pansharpened image X and ground-truth images Y, which is defined as follows:

$$\mathrm{CC} \triangleq \frac{\sum_{i=1}^{w}\sum_{j=1}^{h}(X_{ij}-\mu_X)(Y_{ij}-\mu_Y)}{\sqrt{\sum_{i=1}^{w}\sum_{j=1}^{h}(X_{ij}-\mu_X)^2\sum_{i=1}^{w}\sum_{j=1}^{h}(Y_{ij}-\mu_Y)^2}} \tag{21}$$

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TABLE II
QUANTITATIVE EXPERIMENT RESULTS ON THE RR DATASETS. BEST VALUES ARE HIGHLIGHTED IN RED, AND SECOND BEST VALUES ARE HIGHLIGHT IN BLUE

	GF-2				WV-3				QB			
	CC ↑	SAM ↓	ERGAS ↓	PSNR ↑	CC↑	SAM ↓	ERGAS ↓	PSNR↑	CC↑	SAM ↓	ERGAS ↓	PSNR↑
IHS [12]	0.9358	2.1384	2.6733	34.56	0.8785	6.5765	6.5761	29.70	0.8960	2.0902	1.5189	41.30
Brovey [66]	0.9184	2.6532	2.8273	33.95	0.8695	6.1423	7.2447	29.47	0.9220	1.8312	1.3852	42.25
GS [34]	0.9357	2.1378	2.6577	34.56	0.8875	6.1741	6.4067	30.07	0.9461	1.6626	1.2926	43.36
PCA [14]	0.9279	2.1740	2.7154	34.38	0.8067	6.8444	7.1571	29.26	0.5357	2.0455	2.1416	39.94
SFIM [67]	0.8711	2.3084	3.4159	32.55	0.8150	5.9162	17.6929	28.39	0.8877	1.7027	1.6043	41.44
Wavelet [17]	0.8454	2.5500	3.8334	31.55	0.8060	6.9444	7.8396	27.97	0.8480	2.0921	1.8089	39.57
PNN [7]	0.9774	1.5100	1.3796	40.02	0.7132	27.1191	12.5819	26.05	0.7304	2.2006	2.3872	37.09
PANNET [9]	0.9723	1.6881	1.5759	39.07	0.8058	6.3328	8.0416	28.01	0.8624	1.7674	1.6340	40.64
MSDCNN [10]	0.9688	1.5871	1.5938	38.96	0.7998	21.4707	11.5636	29.84	0.8989	1.6411	1.3108	42.07
GPPNN [25]	0.9743	2.5475	1.1326	41.74	0.9614	5.6678	4.6891	32.97	0.9296	1.6558	1.1503	43.36
SRPPNN [68]	0.9863	1.1370	1.0926	42.17	0.8567	6.7159	7.0963	28.78	0.8283	2.0074	1.8106	39.37
BAM [11]	0.9847	1.1741	1.1426	41.74	0.9450	5.7191	4.2681	33.52	0.8424	1.9655	1.6316	39.89
NormNet [24]	0.9864	1.2162	1.0614	42.36	0.8628	5.0621	19.1811	30.69	0.9422	1.1918	0.9914	45.06
ABFNet	0.9885	1.0215	0.9963	42.91	0.9547	4.6542	3.9742	34.56	0.9701	1.0684	0.7855	47.14

where w and h are the width and height of the images, μ_X and μ_Y denote the mean value of the pansharpened and ground-truth images, and $X_{ij} \in \mathbb{R}^C$ is the spectral vector located at (i, j).

2) **SAM** measures the spectral distortion between the pansharpened and ground-truth images, which can be calculated as follows:

$$\text{SAM} \triangleq \frac{1}{N}\sum_{i=1}^{w}\sum_{j=1}^{h}\arccos\frac{X_{ij}\cdot Y_{ij}}{\|X_{ij}\|_2\cdot\|Y_{ij}\|_2} \tag{22}$$

where $\|*\|_2$ is the Frobenius norm of the spectral vector $*_{ij}$.

3) **ERGAS** measures the overall quality of the pansharpened image, which is given by

$$\text{ERGAS} \triangleq 100R\sqrt{\frac{1}{C}\sum_{k=1}^{C}\frac{\text{RMSE}(B_i)}{M(B_i)}} \tag{23}$$

where R is the upsampling ratio from the LR-MS image to the HR-MS image. $\text{RMSE}(B_i)$ is the root-mean-square error of the ith band between the pansharpened and ground-truth images. C is the number of spectral bands.

4) **PSNR** measures the reconstruction accuracy of the pansharpened image, which is defined as follows:

$$\text{PSNR} = \frac{1}{C}\sum_{k=1}^{C}10\log\frac{2^b-1}{\text{RMSE}(B_i)} \tag{24}$$

where b is the bit depth of the datasets.

2) FR Metrics: They include three metrics that do not require ground truth: D_λ, D_s, and quality of no reference (QNR) [72].

1) $\mathbf{D}_\lambda$ measures the spectral preserving quality between the pansharpened image H and the LR-MS images L as follows:

$$D_\lambda \triangleq \sqrt{\frac{2}{C(C-1)}\sum_{i=1}^{C}\sum_{j=1}^{C}|Q(H_i,H_j)-Q(L_i,L_j)|} \tag{25}$$

where i and j denote the ith and jth band response and Q is the quality index proposed in [73], which measures the structure similarity between the two bands.

2) $\mathbf{D}_s$ measure the spatial consistency between the PAN image P and the pansharpened image by calculating the difference between the H–P image pair and the P–L image pair as follows:

$$D_s \triangleq \sqrt{\sum_{k=1}^{C}|Q(H_i,P)-Q(L_i,P)|}. \tag{26}$$

3) **QNR** combines D_λ and D_s to measure the overall quality of the pansharpened image, which is defined as follows:

$$\text{QNR} \triangleq (1-D_\lambda)(1-D_s). \tag{27}$$

D. Results

We present the experiment results on the RR dataset in Table II, where the best and second best values for each metric are highlighted in red and blue, respectively. Higher values of CC and PSNR and lower values of SAM and ERGAS indicate better performance. It can be seen that the ABFNet achieves superior performance over other methods on the RR dataset. In particular, on the GF-2 and WV-3 datasets, ABFNet shows a remarkable improvement, surpassing other methods by a large margin. On the QB dataset, we still obtain the best results in SAM, ERGAS, and PSNR, and rank second in CC. Notably, our method improves PSNR by 1.29%/3.09%/4.63% on the GF-2/WV-3/QB datasets, respectively. These results demonstrate the effectiveness of our ABFNet on RR pansharpening.

To further illustrate the visual quality of the pansharpened images, we show some examples of the GF-2, WV-3, and QB datasets in Figs. 3–5, respectively. We display both RGB and error maps for each example. The RGB image is generated by stretching and quantizing the red, green, and blue bands of the MS image into [0, 255], which can provide an intuitive impression of the image. However, the RGB image does not reflect the information of other spectral bands or the spectral distortion under the quantization threshold, so we also present

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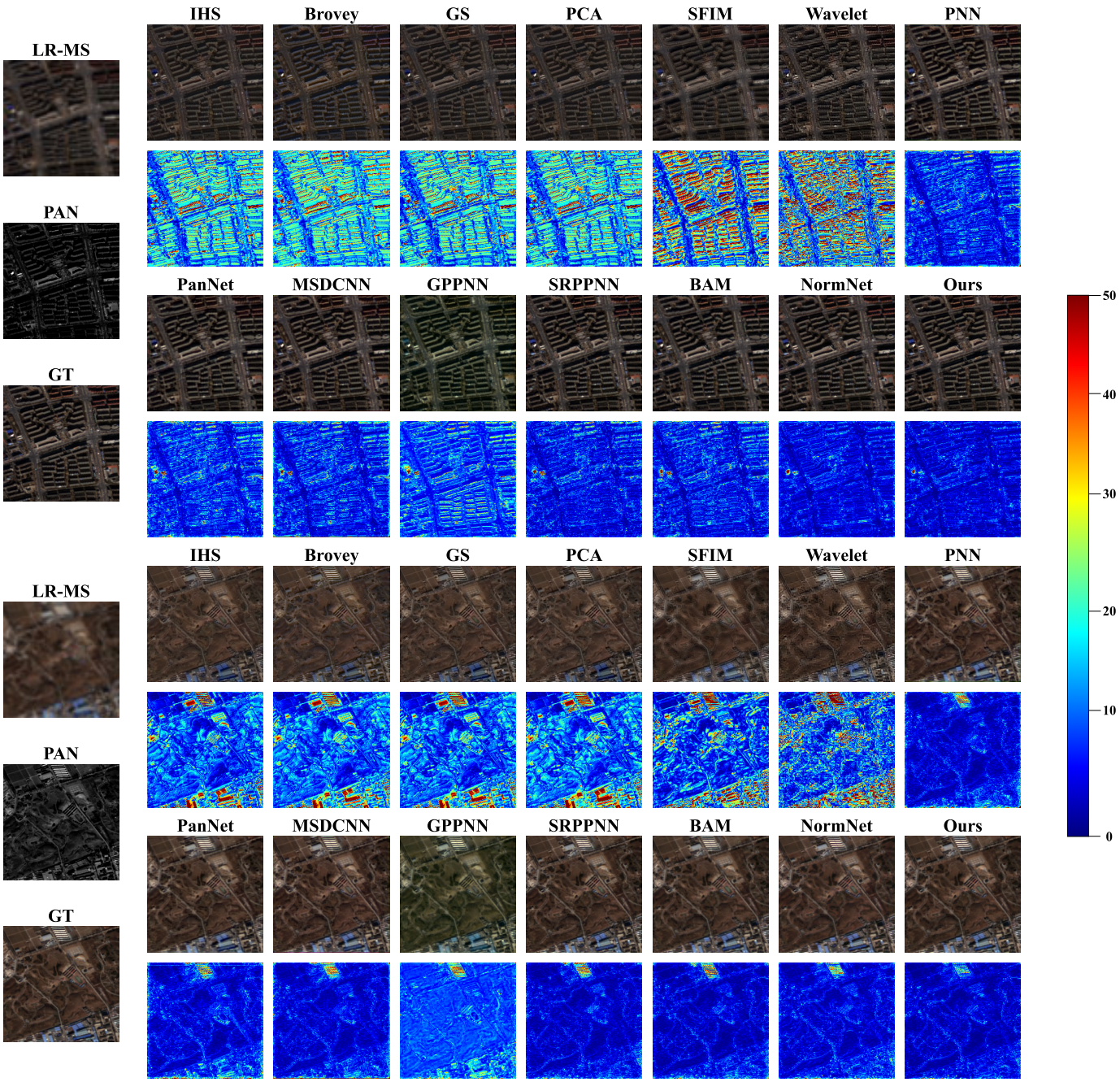


Fig. 3. Pansharpening results of different methods on the GF-2 dataset. The RGB image on the top is generated using red, green, and blue bands of MS images, where the color response is stretched and quantified into $[0, 255]$. The bottom image is the error maps, which calculated the average $L1$ distance between the ground-truth HR-MS image and the generated image. Better viewed with zoomed-in view.

the error maps below the RGB image to indicate the average error of all the spectral bands. The error maps first computes the average $L1$ distance between the generated and ground-truth images and then converts the distance into the heat-map image for easy visualization.

Fig. 3 shows some examples of the GF-2 dataset. All the methods can capture the overall structures with the help of the PAN image. Traditional methods introduce significant color distortions in the generated images due to poor spectral preservation ability. The RGB images of these methods are darker than the ground-truth image, and the error maps show higher errors in repetitive patterns and building roofs. On the other hand, deep-learning-based methods achieve considerable visual improvements over traditional ones, where the RGB images are more realistic and brighter. The error maps display fewer errors with respect to the ground-truth image. However, other deep-learning-based methods still struggle with object edges and junctions of two terrains, but ABFNet can effectively reduce the spectral errors in these areas and faithfully reconstruct the HR-MS image. Fig. 4 shows some examples of the WV-3 dataset. The dataset has eight spectral bands, so we can expect higher errors in the error maps. Traditional

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TABLE III
EXPERIMENT RESULTS ON THE FR DATASETS. BEST VALUES ARE HIGHLIGHTED IN RED, AND SECOND BEST VALUES ARE HIGHLIGHT IN BLUE

	GF-2			WV-3			QB		
	$D_\lambda \downarrow$	$D_s \downarrow$	QNR $\uparrow$	$D_\lambda \downarrow$	$D_s \downarrow$	QNR $\uparrow$	$D_\lambda \downarrow$	$D_s \downarrow$	QNR $\uparrow$
IHS [12]	0.0471	0.2635	0.7020	0.0618	0.2927	0.6643	0.0471	0.2635	0.7020
Brovey [66]	0.0416	0.2567	0.7126	0.0367	0.2782	0.6958	0.0416	0.2567	0.7126
GS [34]	0.0319	0.2520	0.7240	0.0578	0.2908	0.6688	0.0319	0.2520	0.7240
PCA [14]	0.0590	0.2924	0.6665	0.0631	0.1531	0.7976	0.0324	0.2588	0.7173
SFIM [67]	0.0278	0.1078	0.8675	0.0434	0.2095	0.7565	0.0278	0.1078	0.8675
Wavelet [17]	0.0634	0.1191	0.8253	0.0764	0.2227	0.7183	0.0634	0.1191	0.8253
PNN [7]	0.0278	0.2177	0.7609	0.2081	0.1266	0.6918	0.0358	0.1686	0.8018
PANNET [9]	0.0296	0.2116	0.7655	0.0273	0.0265	0.9470	0.0159	0.1235	0.8626
MSDCNN [10]	0.0256	0.2190	0.7612	0.1840	0.1811	0.6687	0.0272	0.1896	0.7884
SRPPNN [68]	0.0129	0.1455	0.8438	0.0761	0.1338	0.8048	0.0358	0.0994	0.8685
GPPNN [25]	0.0321	0.2565	0.7198	0.1926	0.1254	0.7065	0.0883	0.1214	0.8010
BAM [11]	0.0179	0.1758	0.8097	0.0446	0.0863	0.8741	0.0781	0.1240	0.8078
NormNet [24]	0.0231	0.1803	0.8009	0.1053	0.1107	0.7998	0.1053	0.1246	0.7860
Ours	0.0115	0.0954	0.8943	0.0385	0.0867	0.8818	0.0064	0.0801	0.9140

methods produce severe color distortions and blurry RGB images because of low spatial resolution and spectral fidelity. Some deep-learning-based methods, such as PNN, PanNet, and MSDCNN, also suffer from blurriness in their outputs. The proposed ABFNet, on the contrary, can generate plausible HR-MS images that present minimal pixel errors compared with the ground-truth images. The QB dataset has a relatively simple scene, and we can observe better results than the other two datasets in both quantitative metrics in Table II and qualitative examples in Fig. 5. Traditional methods can produce decent results in these scenes, where errors mostly occur at object edges. Deep-learning-based methods exhibit different characteristics, where they show strong noise interference at image boundaries. The boundary artifacts can be clearly observed in both RGB and error maps. However, ABFNet can generate clean results with fewer errors and no boundary artifacts.

The performance of our proposed method and several state-of-the-art methods on three FR datasets (GF-2, WV-3, and QB) is compared in Table III. Lower values of D_λ and D_s indicate better performance, while higher values of QNR indicate better performance. As can be seen from the table, ABFNet consistently surpasses other methods on GF-2 and QB datasets, which demonstrates its superior generation ability from the RR datasets to the FR datasets. On the WV-3 dataset, though ABFNet does not achieve the best metrics, it still ranks second in QNR and third in D_λ and D_s. To further illustrate the quality of the pansharpened HR-MS images, we show some visual examples in Fig. 6. We obtain the RGB images of the LR-MS and HR-MS images by applying the same stretching and quantizing operations on the red, green, and blue bands as in the RR datasets. The HR-MS images exhibit finer details than the LR-MS images, such as the sharpness of the building edges and the richness of the road textures. Moreover, they retain faithful spectral characteristics that are not available in PAN images. These visual results confirm that the ABFNet can effectively enhance the spatial resolution of MS images while preserving their spectral information.

TABLE IV
COMPARISONS OF PARAMETER NUMBER (M) AND FLOPS (G). "PARAM" DENOTES THE PARAMETER NUMBER

	PNN	PanNet	MSDCNN	GPPNN	SRPPNN	BAM	NormNet	ABFNet
Param/K	68.9	68.8	239.0	120.0	898.0	95.1	218.0	80.6
FLOPs/G	4.51	4.51	15.65	5.59	21.85	6.21	13.98	5.26
PSNR/dB	34.39	35.91	36.96	39.36	33.8	38.38	39.37	41.53

We evaluate the trade-off between performance and complexity by comparing the number of parameters and floating point operations (FLOPs) of different methods in Table IV. We also use the average PSNR over the three datasets as a measure of network performance. Our ABFNet only contains 80.6k parameters, only a slight increase in the number of parameters compared with the lightest method PNN [7], which has 68.9k parameters, but we have improved the average PSNR from 34.39 to 41.53 dB. As for the computation complexity, we measure all the networks under the same settings, where the shape of the LR-MS image is $64 \times 64 \times 4$ and the shape of the PAN image is $256 \times 256 \times 1$. The FLOPs of the proposed ABFNet are 5.26G, which is kept at a reasonable level relative to other methods. Notably, compared with NormNet [11], which also uses the adaptive normalization mechanism as the SC module, ABFNet has reduced the parameters from 218k to 80.6k, and FLOPs from 13.98G to 5.26G, but increased the PSNR by 3.15 dB.

Overall, these results show that the asymmetric bidirectional fusion paradigm, which transfers complementary features in both directions, is effective and efficient for multispectral pansharpening.

E. Ablation Study

To demonstrate the effectiveness of the proposed asymmetric bidirectional fusion paradigm, two ablation experiments are conducted.

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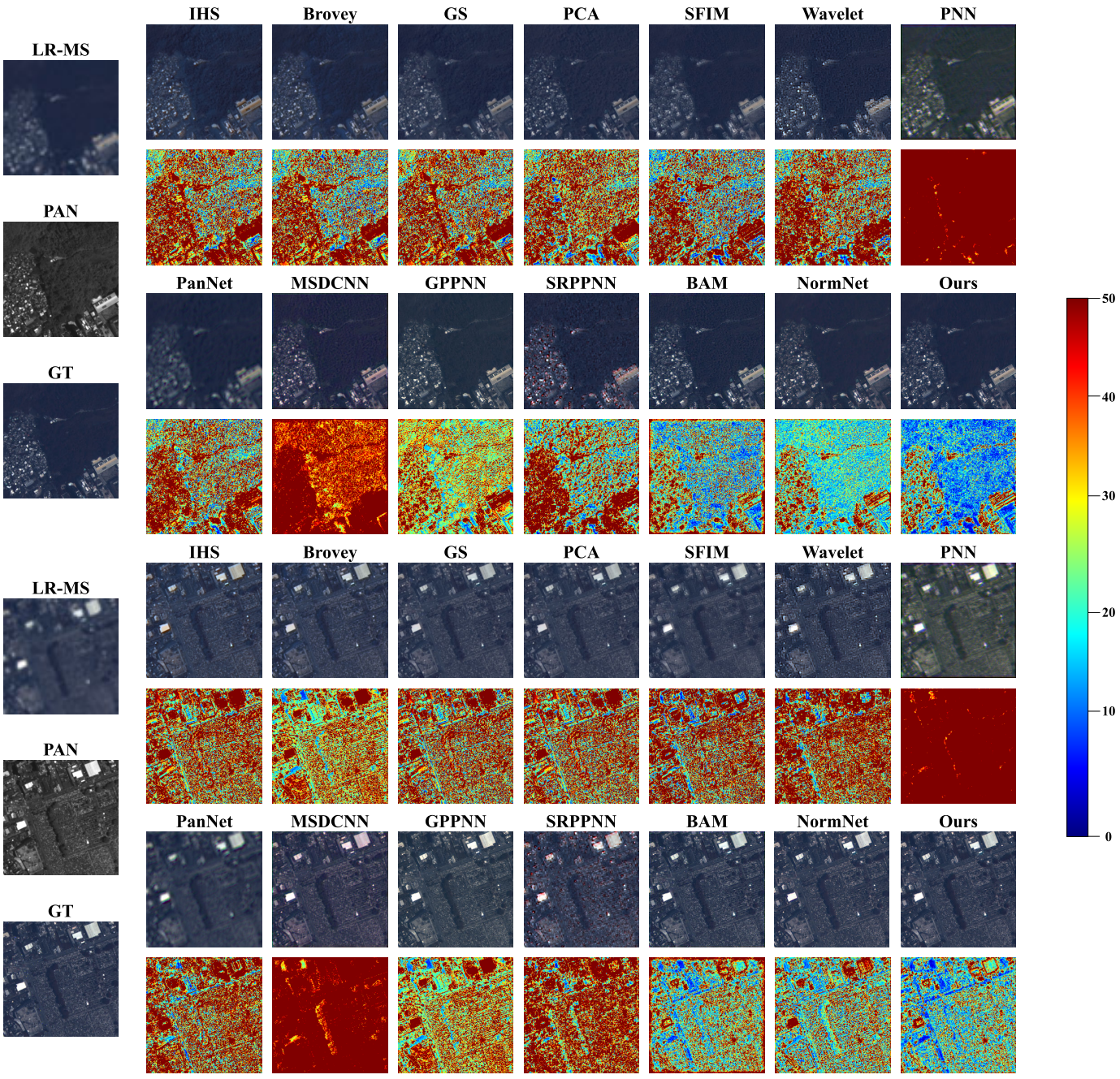


Fig. 4. Pansharpening results of different methods on the WV-3 dataset. The RGB image on the top is generated using red, green, and blue bands of MS images, where the color response is stretched and quantified into $[0, 255]$. The bottom image is the error maps, which calculated the average $L1$ distance between the ground-truth HR-MS image and the generated image. Better viewed with zoomed-in view.

TABLE V

ABLATION EXPERIMENTS ON THE BIDIRECTIONAL FUSION PARADIGM

SC	SRC	GF-2				WV-3				QB			
		CC	SAM	ERGAS	PSNR	CC	SAM	ERGAS	PSNR	CC	SAM	ERGAS	PSNR
×	✓	0.9797	1.3504	1.3402	40.53	0.9416	5.9971	4.4070	33.54	0.9427	1.4898	1.1963	44.09
✓	×	0.9872	1.0383	1.0894	42.30	0.9481	4.6419	4.5753	33.20	0.9682	1.0533	0.8252	46.76
✓	✓	**0.9885**	**1.0215**	**0.9963**	**42.91**	**0.9547**	**4.6542**	**3.9742**	**34.56**	**0.9701**	**1.0684**	**0.7855**	**47.14**

1) Bidirectional Fusion: It is a key component of our proposed model, as it allows the two images (PAN and LR-MS) to exchange and integrate complementary information through the SC and SRC modules. To demonstrate the effectiveness of this mechanism, we conduct ablation experiments using unidirectional fusion, which only adopts one

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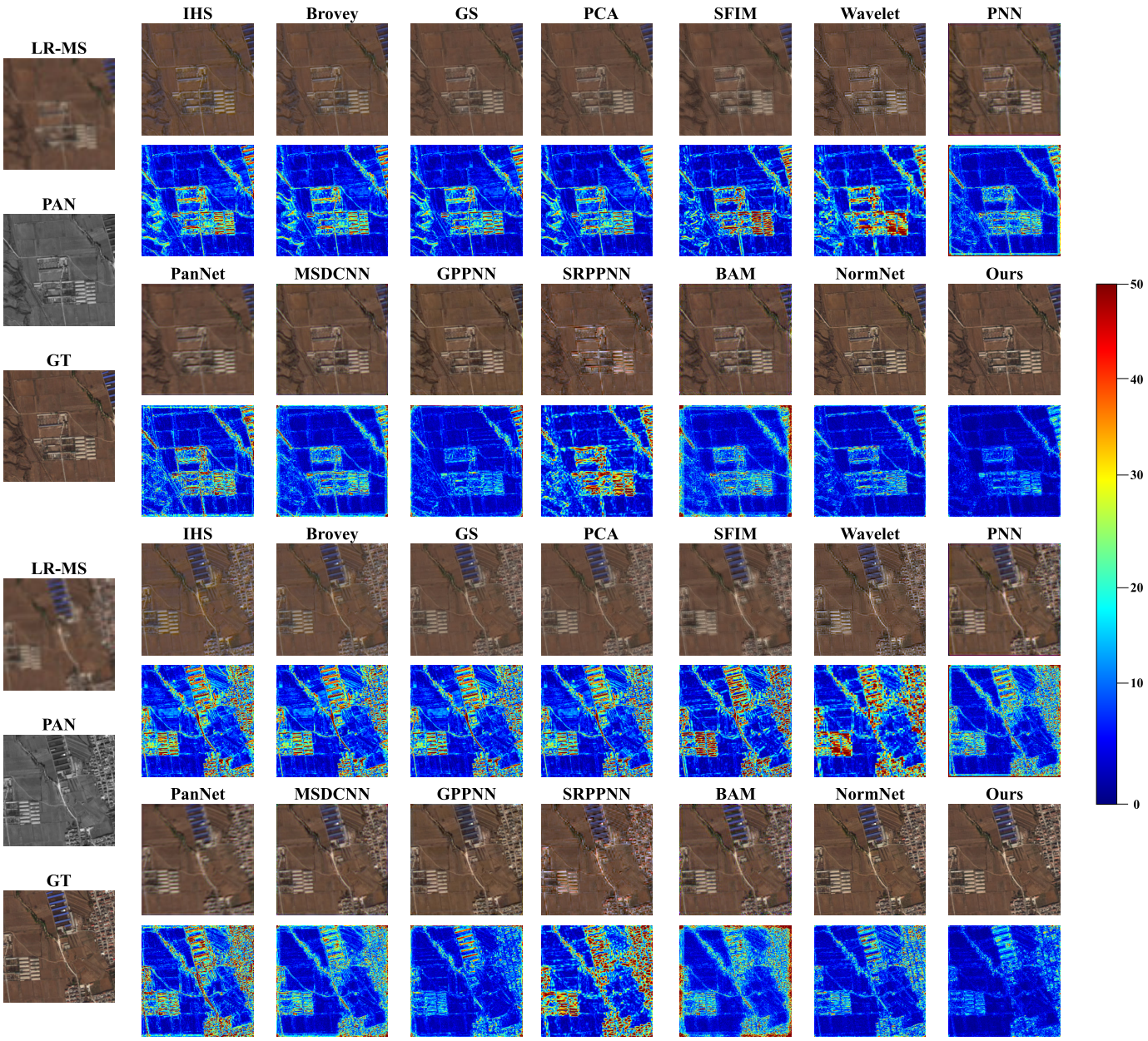


Fig. 5. Pansharpening results of different methods on the QB dataset. The RGB image on the top is generated using red, green, and blue bands of MS images, where the color response is stretched and quantified into [0, 255]. The bottom image is the error maps, which calculated the average $L1$ distance between the ground-truth HR-MS image and the generated image. Better viewed with zoomed-in view.

TABLE VI
ABLATION EXPERIMENTS ON THE ASYMMETRIC FUSION MODULES

	GF-2				WV-3				QB			
	CC	SAM	ERGAS	PSNR	CC	SAM	ERGAS	PSNR	CC	SAM	ERGAS	PSNR
SC-SC	0.9743	1.1363	7.7705	40.15	0.7161	5.7053	53.9172	25.11	0.9478	1.1957	1.1283	45.23
SRC-SRC	0.9756	1.2009	5.3757	40.95	0.9529	4.3766	4.1323	34.33	0.9687	1.0482	0.8138	47.03
SC-SRC	**0.9885**	**1.0215**	**0.9963**	**42.91**	**0.9547**	**4.6542**	**3.9742**	**34.56**	**0.9701**	**1.0684**	**0.7855**	**47.14**

of the two modules and performs fusion in one direction, either from PAN to LR-MS (SRC module) or from LR-MS to PAN (SC module). The results in Table V show that the model's performance drops significantly when unidirectional fusion is used, as indicated by the lower values of PSNR in the three datasets. This confirms that the bidirectional fusion mechanism enhances the feature fusion process and improves the quality of the reconstructed HR-MS image.

2) Asymmetric Modules: They are another important aspect of the ABFNet, as they exploit distinct fusion modules for different types of information. To verify the necessity of this

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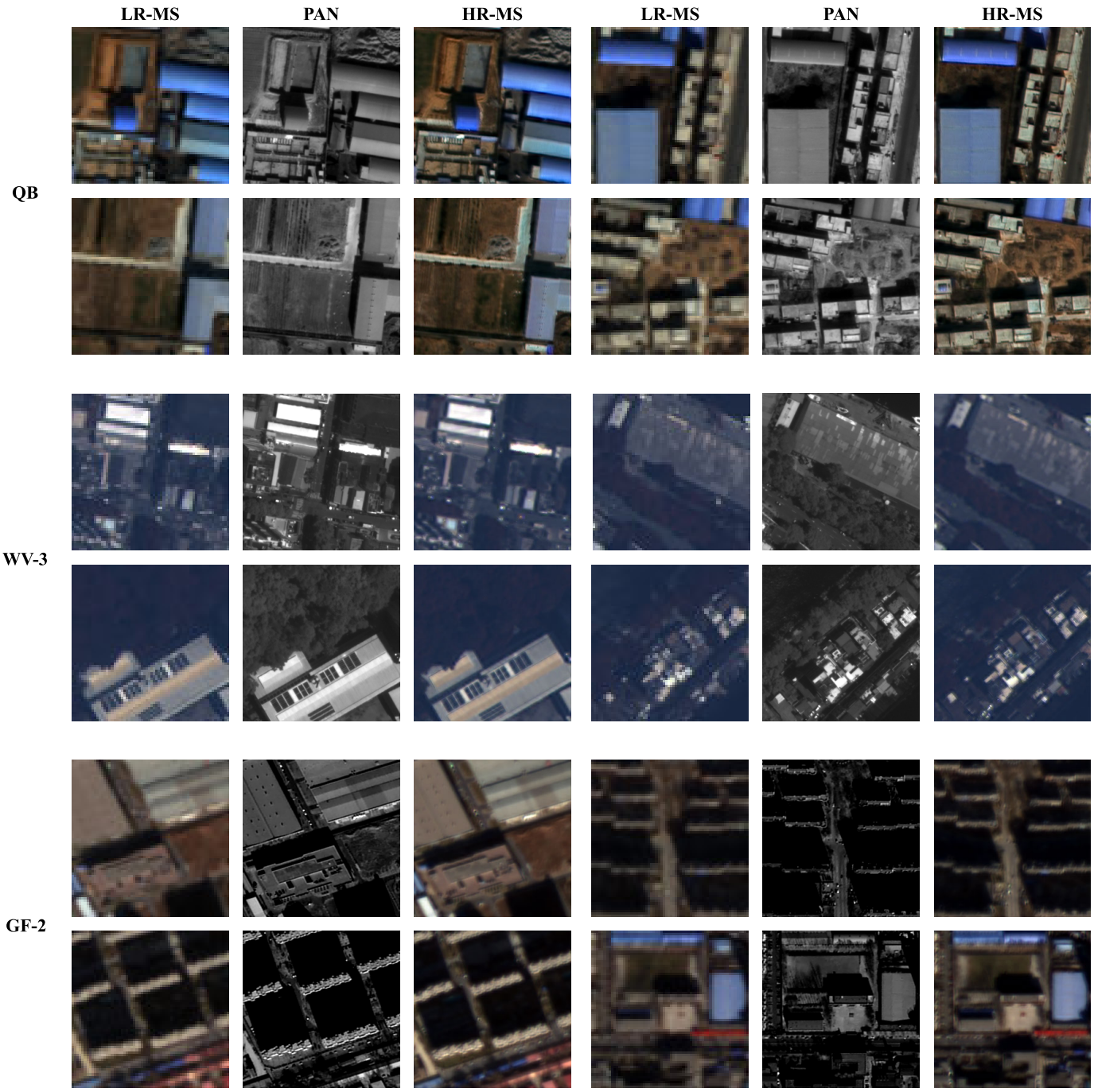


Fig. 6. Examples of pansharpening results on the FR datasets.

design, we conduct ablation experiments using symmetric fusion modules, where the bidirectional fusion modules are identical, and compare the results with those of the asymmetric fusion modules. The results are shown in Table VI. We can see that when both fusion modules are set to either the SRC blocks (SRC–SRC) or the SC blocks (SC–SC), the performance of the model deteriorates significantly, as evidenced by the higher values of SAM and ERGAS. The results suggest that using symmetric fusion modules cannot capture the diverse and complementary information from the MS and PAN features. But, by adopting the asymmetric modules, the network can extract more relevant and richer features, leading to more realistic and accurate results.

3) Effect of Normalization in SC Module: The IN in the SC module normalizes the PAN features based on its per-dim mean and standard deviation. To further evaluate its effects, we conduct ablation experiments without the IN layer, and the results are presented in Table VII. Without the IN layer, the performance drops on all the evaluation metrics, which demonstrates that the IN layer can help the ABFNet generate better HR-MS images. To visually demonstrate the effect of the IN layer, we visualize the output PAN features of the

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TABLE VII
ABLATION EXPERIMENTS ON THE IN LAYER

	GF-2				WV-3				QB			
	CC	SAM	ERGAS	PSNR	CC	SAM	ERGAS	PSNR	CC	SAM	ERGAS	PSNR
w/o IN	0.9864	1.1059	1.1259	41.99	0.9403	4.8331	4.1498	33.73	0.9653	1.1489	0.8628	46.46
w/ IN	**0.9885**	**1.0215**	**0.9963**	**42.91**	**0.9547**	**4.6542**	**3.9742**	**34.56**	**0.9701**	**1.0684**	**0.7855**	**47.14**

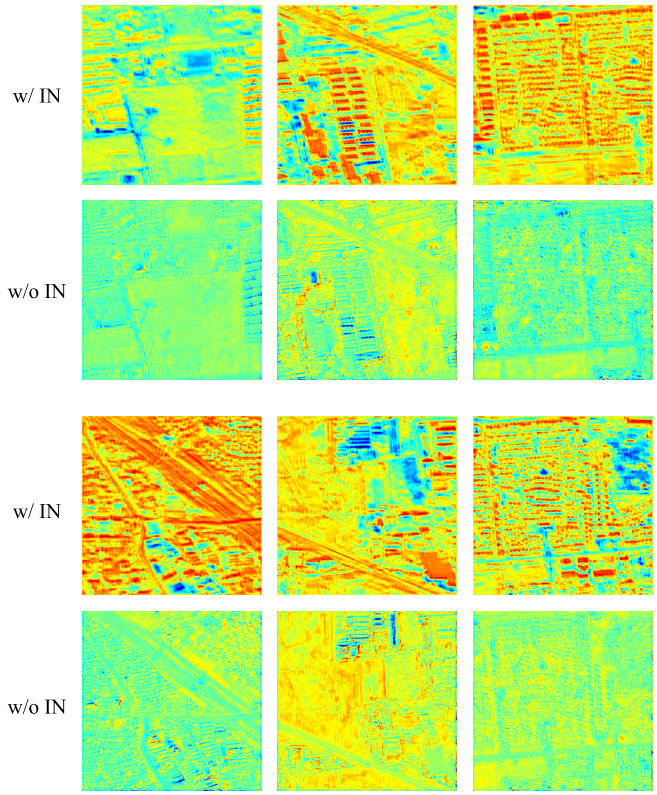


Fig. 7. Visualization of the final PAN features. "w/ IN" means network with IN layer, and "w/o IN" means network without IN layer.

last BB as Fig. 7. With the IN layer, the PAN features enjoy high feature contrast, which can distinguish different semantic objects. On the contrary, the PAN features produced by the network without IN layer convert less semantic information, where boundaries are blurred due to closer pixel intensities. The visualization results indicate that by adopting the IN layer to standardize PAN features, the SC module can better recalibrate the features' distribution and reinforce the PAN features with plausible spectral properties.

V. LIMITATIONS AND FUTURE WORKS

The ABFNet can achieve the state-of-the-arts performance and generate plausible HR-MS images on various satellite datasets and across different spatial resolutions. It still has the following limitations.

First, as can be visualized in Fig. 6, some HR-MS images blur the details of the PAN image to match the color distribution of the LR-MS image. For example, the HR-MS image on the GF-2 dataset blurs the roofs, which makes the chimney hard to recognize. We conjecture that it may be caused by the $L1$ loss and may alleviate it by adding adversary and perception loss as in [18], which will be left for future works.

Second, the SC and SRC modules fuse the two features in a point-to-point manner, which mainly operates on the channel dimension of the PAN and LR-MS features. This restricted the interpretability of the fusion modules on the spatial dimension. To improve the interpretability of the models, spatial attention [74] and deformable convolution [75] can be incorporated into the fusion modules.

VI. CONCLUSION

In this article, we propose a novel and effective network for remote sensing pansharpening, which is the task of generating HR-MS images by fusing LR-MS images and high-resolution PAN images. The proposed network, called the ABFNet, uses a bidirectional fusion pipeline that enables reciprocal feature learning and fusion between the PAN and LR-MS images. Moreover, we propose two asymmetric fusion modules that adapt to the different properties and demands of the heterogeneous PAN and LR-MS features. The asymmetric fusion modules can integrate spatial details and spectral information in a balanced and efficient way. By progressively reinforcing the PAN and LR-MS features, the ABFNet can generate HR-MS images that outperform the existing methods in terms of both visual quality and objective metrics.

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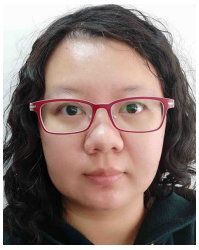
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